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Master's thesis

Topological quantum computing with the Kitaev chain

Solving quantum computing and
providing the world with infinite porridge

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Abstract

This thesis begins by reviewing Kitaevs original article on the Kitaev chain [1], a one-dimensional superconducting chain of spinless fermions. We outline solutions of this model where Majorana zero modes arise at the edges, providing a basis for topologically protected qubits. Further we explore how the Kitaev chain can be used as a qubit, specifically studying the T-chain geometry, showing that this system can implement the Pauli-X and Pauli-Y quantum gates by braiding of Majoranas. But we also show that this system does not provide the basis for a complete quantum computer, and discuss the missing operations.

We explore methods of numerical simulations of the model, implementing both an adiabatic method allowing for truncation of higher energy states in hopes of speeding up numerical simulations while maintaining accuracy, and direct diagonalization of the full spectrum (for comparison). Initial testing proves to replicate the results of a previous student for the case of cutting a chain, an operation in which a single chain starts with all sites in the topologically non-trivial phase, and then gradually turning the hopping and pairing terms of the central bond to zero, before gradually turning them on again. Further we simulate the Pauli-X and -Y gates, in which two different pairs of majoranas are exchanged, showing that the ground state degeneracy is preserved throughout the operation. For the numerical results we specifically discuss the evolution of the ground state energies for varying total times of the operation T, varying levels of truncation, and the results are heavily restricted in the total number of sites on the chain, at most reaching $N = 13$ sites for a simulation. [Her vil jeg legge inn resultater].

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Chapter 1

The Kitaev chain

The Kitaev chain is a toy model of a one dimensional spinless p-wave superconductor, and is an example of material that can realize a topological superconducting phase. It consists of a chain of spinless fermions with nearest-neighbour hopping and superconducting pairing, under the influence of a chemical potential. The Hamiltonian takes the form

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{1.1}$$

where a_j and $a_j^\dagger$ are annihilation and creation operators for the fermionic sites $j = 1, 2, ..., L$ in the chain, where L (sometimes N) is the total amount of sites. ω is a hopping amplitude, μ is the chemical potential, and $\Delta = |\Delta| e^{i\theta}$ is the induced superconducting gap. The Hamiltonian serves to give rise to zero energy Majorana fermion quasiparticles at its boundaries. Simply put, these Majorana fermions are theorized to be useful as qubits, and a network of chains can be used for quantum computation.

Majorana fermions are interesting candidates for qubits as their topological nature provides robustness against local perturbations and certain sources of decoherence. Their non-Abelian nature allows for braiding, where the majorana fermions exchange position and information about the path is stored in the wavefunction. This is the core of the process that allows computation to take place, and we will study this closer in a later chapter **??**.

In the first two sections 1.1, 1.2 (and this heading) of this chapter we closely follow the original theory laid out by Kitaev [1], re-deriving the main results and filling in with intermediate steps and discussion for clarity. More specifically we cover the origin of the Majorana fermions and the phases of the Kitaev chain. We also study BdG formalism and apply it to the Kitaev chain as parts of the phase discussion is based on these results. Further the BdG formalism can also offer great reductions in the dimensions of the problem greatly, and is as such a common formalism throughout the field.

1.1 Majorana fermions

A Majorana fermion is a particle that is its own antiparticle. The Majorana operators can be defined as

$$c_{2j-1} = a_j + a_j^\dagger, \quad c_{2j} = \frac{a_j - a_j^\dagger}{i}, \quad (j = 1, ..., N) \tag{1.2}$$

which are hermitian, and anticommute, meaning they satisfy the relations

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = c_l c_m + c_m c_l = 2\delta_{lm}, \quad (l, m = 1, ..., 2N). \tag{1.3}$$

Using the definitions of the majorana operators 1.2 we can find the number operator for a site j

$$a_j^\dagger a_j = \frac{1}{2}\left(1 + i c_{2j-1} c_{2j}\right), \tag{1.4}$$

where the term $ic_{2j-1}c_{2j}$ corresponds to a phase error. If c_{2j-1} and c_{2j} are not on the same site, but separated in space, then $c_{2j-1}c_{2j}$ would represent an interaction between two separated Majorana sites, requiring the environment to couple the Majorana fermions simultaneously to create the phase error. Physical noise and imperfections are typically local, so placing the Majorana sites far enough away from each other would decrease the risk of a phase error. Further, a single Majorana operator cannot appear by itself in a reasonable Hamiltonian as it would not preserve fermionic parity; flipping the system from an even to odd (or vice versa) number of fermions. As such, the Majorana operators will always come in pairs $c_i c_l$, and thus no parity preserving term can involve just one isolated Majorana. Together this tells us that errors would either have to break parity, or couple both separated Majoranas at once, which is highly unlikely and thus the Majorana fermions are protected from local errors.

1.1.1 Rewriting the Hamiltonian in terms of Majorana operators

We can express the Hamiltonian in terms of Majorana operators. As we will see, this transformation will provide insight about the structure of the phases and the emergence of the edge modes. Before doing so we simplify the calculation by hiding the phase θ in the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{1.5}$$

$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{1.6}$$

We can now perform the transformation of the Hamiltonian by inverting these relations and inserting into the original Hamiltonian 1.1. Together with the hermiticity and anticommutation relations 1.3 we get the Hamiltonian

$$H = \frac{i}{2}\sum_j \left[-\mu c_{2j-1}c_{2j} + (\omega + |\Delta|)\, c_{2j}c_{2j+1} + (-\omega + |\Delta|)\, c_{2j-1}c_{2j+2}\right]. \tag{1.7}$$

This calculation is shown in appendix A.1. Staring at this Hamiltonian, we can first see that it is on the general quadratic Majorana form

$$H = \frac{i}{4}\sum_{l,m} A_{lm} c_l c_m, \quad A^T = -A \tag{1.8}$$

with A being a real and antisymmetric matrix, encoding the intrinsic particle-hole symmetry of the superconducting system. The single particle spectrum is fully determined by the eigenvalues of A, reducing the problem from a 2^L-dimensional many-body operator problem in Fock space to a single-particle linear algebra problem on a $2L \times 2L$ matrix. This is the basis of BdG, as we will get back to later. Further, we see that all direct mentions of the complex phase θ are gone from H. This reflects that θ

can be removed by an appropriate gauge transformation, and is therefore irrelevant for the bulk spectrum and the topological classification of the phases, which we will also touch upon later.

The Majorana representation also gives us a geometric picture of the chain. The Hamiltonian now consists of three terms, providing three different types of couplings. An on-site term $-\mu c_{2j-1}c_{2j}$, and two nearest-neighbour terms $(\omega + |\Delta|)c_{2j}c_{2j+1}$ and $(-\omega + |\Delta|)c_{2j-1}c_{2j+2}$. The on-site term couples two Majorana modes per site j, so for a chain of length L we get $2L$ Majorana operators. We can garner further insight by studying specific cases for the parameters μ, ω and $|\Delta|$ of the Hamiltonian.

Topologically trivial case $\omega = |\Delta| = 0, \mu > 0$

In the case where μ dominates the Hamiltonian reduces to

$$H = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) = \frac{i}{2} \mu \sum_j c_{2j-1} c_{2j}. \tag{1.9}$$

In this case the system is topologically trivial as the Majoranas on each site are glued together to form a complex fermion. There are no unpaired Majorana fermions at the edges of the chain, making this case uninteresting for further study.

Topologically non-trivial case $\omega = |\Delta| > 0, \mu = 0$

In the case where the hopping and superconducting parameters dominate the Hamiltonian reduces to

$$H = i \sum_j \omega c_{2j} c_{2j+1}, \tag{1.10}$$

where the nearest neighbours between different sites dominate. This is an interesting case as c_2 couples to c_3, c_4 couples to c_5, and so on. But c_1 and c_{2L}, the Majorana edge states, do not appear in the Hamiltonian. They have however not disappeared physically, but rather remain as zero-energy Majorana modes localized at the edges of the wire. Together they form a nonlocal fermionic zero mode. This is the most important insight from the transformation of the Hamiltonian as these are the states we were looking to create.

We can now define new annihilation and creation operators spanning the sites j and $j+1$

$$d_j = \frac{1}{2} \left(c_{2j} + i c_{2j+1} \right) \tag{1.11}$$

$$d_j^\dagger = \frac{1}{2} \left(c_{2j} - i c_{2j+1} \right). \tag{1.12}$$

Such that we can rewrite the Hamiltonian

$$H = 2\omega \sum_{j=1} \left(f_j^\dagger f_j - \frac{1}{2} \right). \tag{1.13}$$

As the edge states are not mentioned in the Hamiltonian, this expression fully determines the energy spectrum of the bulk. We see that the bulk consists of fermionic modes with energy $\propto 2\omega$. Further, the ground states satisfy the condition $d_j|\psi\rangle = 0$ for all j. As such, all these modes will remain empty, and any ground state $|\psi\rangle$ is fully determined by

the edge majoranas c_1 and c_{2L}. Therefore we get two orthogonal states where the above ground state condition for the bulk is true; $|\psi_0\rangle$ and $|\psi_1\rangle$. These ground states will be eigenstates of the hermitian operator $-ic_1c_{2L}$ as the edge Majoranas represent unpaired degrees of freedom. This operator also squares to 1, and as such has eigenvalues ± 1. We get

$$-ic_1c_{2L}|\psi_0\rangle = |\psi_0\rangle, \quad -ic_1c_{2L}|\psi_1\rangle = -|\psi_1\rangle, \tag{1.14}$$

where the labeling of the states were chosen arbitrarily. We see that the total fermion parity operator

$$P = \prod_j (-ic_{2j-1}c_{2j}) \tag{1.15}$$

has the same contribution from the bulk for both ground states. So the difference in parity is solely a result of the edge contribution $-ic_1c_{2L}$. The even parity of $|\psi_0\rangle$ represents the state being a superposition of states with an even number of electrons. And $|\psi_1\rangle$ has an odd parity, representing an odd number of electrons. Together they reflect the boundary degrees of freedom from the unpaired Majorana modes at the edges.

Arbitrary values of ω, μ, Δ

For arbitrary values of ω, μ, Δ the Hamiltonian has the general quadratic form (1.8) in terms of the Majorana operators. This means the Hamiltonian can be diagonalized into independent fermionic modes by a change of basis, implemented by a matrix W. We require that W is a $2N \times 2N$ real orthogonal ($W^TW = WW^T = I$) matrix, and that it has two further properties; the first being that it defines new operators $\{b'_m, b''_m\}$ by the linear transformation $b = Wc$, or

$$\begin{pmatrix} b'_1 \\ b''_1 \\ \vdots \\ b'_N \\ b''_N \end{pmatrix} = W \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{2N-1} \\ c_{2N} \end{pmatrix}. \tag{1.16}$$

This property is ensured by W being real and orthogonal, such that after the transformation the Majorana anticommutation relations are preserved. This can be shown using $b_\gamma = \sum_\alpha W_{\gamma\alpha}c_\alpha$, where $W_{\gamma\alpha}$ are just real numbers, meaning they commute

$$\{b_\gamma, b_\delta\} = \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}\{c_\alpha, c_\beta\} \tag{1.17}$$

$$= \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}2\delta_{\alpha\beta} \tag{1.18}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W_{\delta\alpha} \tag{1.19}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W^T_{\alpha\delta} \tag{1.20}$$

$$= 2\delta_{\gamma\delta} \tag{1.21}$$

where the last step is a result of W being orthogonal. The second property stems from the antisymmetry of A, such that in this basis A acquires a canonical block-diagonal

form

$$WAW^T = \begin{pmatrix} 0 & \epsilon_1 & & & \\ -\epsilon_1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & \epsilon_N \\ & & & -\epsilon_N & 0 \end{pmatrix}. \tag{1.22}$$

This condition stems from a mathematical theorem on the canonical form of real antisymmetric matrices, which states that if we have a real antisymmetric matrix A of size $2N \times 2N$ there exists a real orthogonal matrix W such that A can be transformed into a block-diagonal form

$$WAW^T = \text{diag}(C_1, C_2, \cdots, C_N), \quad C_j = \begin{pmatrix} 0 & -\lambda_j \\ \lambda_j & 0 \end{pmatrix} \tag{1.23}$$

with $\lambda_j > 0$ being real numbers, stemming from the pairs of complex eigenvalues $\pm i\lambda_j$ for A. This theorem is a standard result in matrix analysis, and a proof can be found in [2] [1]. Once we have a matrix W that satisfies the above conditions we can reduce the Hamiltonian further to a canonical form by inverting the transformation $b_\gamma = \sum_\alpha W_{\gamma\alpha} c_\alpha$ to $c_\alpha = \sum_\gamma W^T_{\alpha\gamma} b_\gamma = \sum_\gamma W_{\gamma\alpha} b_\gamma$

$$\begin{aligned} H_{\text{canonical}} &= \frac{i}{4} \sum_{\alpha,\beta} A_{\alpha\beta} c_\alpha c_\beta && (1.24)\\ &= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} A_{\alpha\beta} W_{\gamma\alpha} b_\gamma W_{\delta\beta} b_\delta && (1.25)\\ &= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} W_{\gamma\alpha} A_{\alpha\beta} W_{\delta\beta} b_\gamma b_\delta && (1.26)\\ &= \frac{i}{4} \sum_{\gamma,\delta} \left(WAW^T \right)_{\gamma\delta} b_\gamma b_\delta && (1.27)\\ &= \frac{i}{4} \sum_{m=1}^{N} \begin{pmatrix} b'_m & b''_m \end{pmatrix} \begin{pmatrix} 0 & \epsilon_m \\ -\epsilon_m & 0 \end{pmatrix} \begin{pmatrix} b'_m \\ b''_m \end{pmatrix} && (1.28)\\ &= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m \left(b'_m b''_m - b''_m b'_m \right) && (1.29)\\ &= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m 2 b'_m b''_m && (1.30)\\ &= \frac{i}{2} \sum_{m=1}^{N} \epsilon_m b'_m b''_m = \sum_{m=1}^{N} \epsilon_m \left(\tilde{a}^\dagger_m \tilde{a}_m - \frac{1}{2} \right) \quad (\epsilon_m \geq 0). && (1.31) \end{aligned}$$

Where we used that $\{b'_m, b''_m\} = 0$ as they are separate components of b. In this form b'_m and b''_m are real linear combinations of c_{2j-1} and c_{2j}, retaining the commutation relations. N is the number of fermionic sites in the system, here $N = L$. The complex fermion quasiparticles are defined as $\tilde{a}_m = \frac{1}{2}\left(b'_m + i b''_m\right)$ and $\tilde{a}^\dagger_m = \frac{1}{2}\left(b'_m - i b''_m\right)$.

The key take-away here is that the Kitaev chain Hamiltonian, due to being quadratic, can be diagonalized into independent quasi particles $\tilde{a}_m$ with energies ϵ_m. This is how

[1] There are apparently many books on this topic, none of which i had access to. This is the best source i could find without proving it directly.

the problem can be reduced from a 2^L-dimensional many-body problem to single-particle energies and modes on a $2L \times 2L$ matrix using BdG formalism as we will later cover 1.3.

1.1.2 Properties of Majorana fermions

As the previous subsection also started by stating, a majorana fermion is a particle that is its own antiparticle. In other words, it obeys

$$c = c^\dagger. \tag{1.32}$$

At first glance[2] it might appear to be a fairly tame statement. This is however an incredibly exciting statement which gives the Majorana fermions exciting properties. This equation essentially enforces that applying a majorana fermion annihilation operator is the same as applying a creation operator: removing a majorana fermion is identical to creating one.

"an unpaired Majorana at a topologically nontrivial defect must have a counterpart somewhere else in the system" [3]

Superconductors - particle hole symmetry

"One peculiar thing about Majorana “fermions” is that no state is annihilated by them $\gamma i|\psi\rangle i = 0 \rightarrow 0 = \gamma 2i|\psi i = |\psi i$. There is no notion of the occupancy of a Majorana state. Asking whether a Majorana fermion is occupied is akin to asking whether the dog has Buddha-nature" - *https* : *//nbi.ku.dk/english/theses/masters − theses/asbjoern − rasmussen/Asbj*$_r$*nRasmussen.pdf*

What happens to the schrodinger equation? E = 0

Nonlocality?

1.2 Phases of the Kitaev chain

The Kitaev chain is really only interesting to us due to its promise of Majorana fermions. We have previously seen that these quasiparticles only appear for specific parameters. In other words the Kitaev chain can host several phases where the Majorana fermions are, or are not present. Previously we have referred to these phases as the topologically non-trivial and the topologically trivial phase, respectively. The goal of this section is to further our understanding of which relation of parameters cause the chain to be in which phase, with a specific focus on the distinction between the topological and the trivial phase.

1.2.1 Phases of the bulk spectrum

The bulk spectrum for energy vs. momentum q for a chain of infinite length is given by

$$\epsilon(q) = \pm\sqrt{(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q)}, \quad -\pi \leq q \leq \pi. \tag{1.33}$$

Finding this relation is done by Fourier transforming the Hamiltonian to momentum space as shown in appendix A.2 and then finding its eigenvalues, as we will show in 1.3. This was done using periodic boundary conditions as we expect the same result when $L \rightarrow \infty$ for a chain. For now you can take this for granted as its derivation gives little insight into the physics of the system. This energy spectrum will tell us where the

[2] In fact, 6 months of first glances for some

system is gapped ($\epsilon(q) \neq 0$), and where gap closings ($\epsilon(q) = 0$) occur. The gap closings mark boundaries between topological phases [3]. We now find the gap closings

$$\epsilon(q) = 0 \tag{1.34}$$

$$\pm\sqrt{(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q)} = 0 \tag{1.35}$$

$$(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q) = 0 \tag{1.36}$$

Both terms in this equation are real values squared, and must therefore be nonnegative. They must also both individually vanish to fulfill the equation, and we thus get two equations

$$2\omega\cos(q) + \mu = 0, \quad \sin(q) = 0. \tag{1.37}$$

From the second equation we get $\sin(q) = 0$ for $q = 0, \pi$. Plugging this into the first equation we get

$$\begin{aligned} 2\omega\cos(0) + \mu = 0 \qquad & \qquad 2\omega\cos(\pi) + \mu = 0 \\ 2\omega = -\mu \qquad & \qquad 2\omega = \mu \end{aligned} \tag{1.38}$$

Meaning the bulk gap closes when $2|\omega| = |\mu|$, defining a phase boundary between the topologically trivial and non-trivial superconducting phases.

From our previous findings we know that the system will be in the topologically non-trivial phase when $\mu < \omega$ (from the case $\omega = |\Delta| > 0, \mu = 0$), which is in the domain $|\mu| < 2|\omega|$. We here expect that if the parameters change without crossing the bulk gap boundary $2|\omega| = |\mu|$ the system should remain in the same phase. Further we need $\Delta \neq 0$ to remain in the superconducting phase. Meaning the phases of the system can be divided into

$2|\omega| > |\mu|, \Delta \neq 0$: Topologically non-trivial superconductor (Majorana edge states).
$2|\omega| = |\mu|, \Delta \neq 0$: The superconducting bulk gap closes (phase boundary).
$2|\omega| < |\mu|, \Delta \neq 0$: Topologically trivial superconductor (no Majorana edge states).
$2|\omega| > |\mu|, \Delta = 0$: Gapless metal.
$2|\omega| < |\mu|, \Delta = 0$: Gapped insulator.

1.2.2 Boundary modes

In the previous subsection we expected the phase of the system to remain the same as long as the bulk gap boundary is not crossed. We can check that this is true by studying the boundary modes, as they correspond to eigenvectors of A localized near the ends of the chain, or in other words can confirm whether or not the phase is in the topological phase containing Majorana fermions. Such zero modes should be real linear combinations of c_{2j-1}, c_{2j}, having the form

$$b' = \sum_j \beta'_j c_{2j-1}, \quad b'' = \sum_j \beta''_j c_{2j} \tag{1.39}$$

where b' and b'' should be localized on opposite sides of the chain. Here β'_j, β''_j are real coefficients of the Majorana operators. If b' and b'' are zero modes they have no energy and should commute with the Hamiltonian $[H, b'] = [H, b''] = 0$. Further, because

[3] For a more detailed dive into how the closings of the bulk gap signify topological phase transitions see Sec. II.A in [4]

the Hamiltonian only couples nearest neighbours the matrix A will be tridiagonal with general values a, b, c, meaning the bulk (excluding the edge modes) can be expressed by

$$a'\beta'_{j+1} + b'\beta'_j + c'\beta'_{j-1} = 0, \tag{1.40}$$
$$a''\beta''_{j+1} + b''\beta''_j + c''\beta''_{j-1} = 0. \tag{1.41}$$

The values a, b, c for their respective β will follow from the Hamiltonian; b being the on-site term μ, while a and c will be the off-diagonal pairing terms $(\omega \pm |\Delta|)$. These follow from the commutation relations mentioned above, and will allow us to find solutions for b', b''. We will here show the calculation for b', and show the final result of b'', which has a similar calculation.

$$[H, b'] = \sum_{l=1}^{L} \beta'_j \, [H, c_{2l-1}] \tag{1.42}$$
$$= \sum_{l=1}^{L} \beta'_j \left(i\mu\delta_{jl} c_{2j} + i(\omega + |\Delta|)\delta_{l,j+1} c_{2j} - i(-\omega + |\Delta|)\delta_{lj} c_{2j+2} \right) \tag{1.43}$$
$$= i\sum_{j=1}^{L} \mu\beta'_j c_{2j} + i\sum_{j=1}^{L-1} \beta'_{j+1}(\omega + |\Delta|)c_{2j} - \sum_{j=2}^{L-1} \beta'_{j-1}(-\omega + |\Delta|)c_{2j} \tag{1.44}$$
$$= 0 \tag{1.45}$$
$$[H, b''] = -i\sum_{j=1}^{L} \mu\beta''_j c_{2j-1} - i\sum_{j=2}^{L-1} \beta''_{j-1}(\omega + |\Delta|)c_{2j-1} + i\sum_{j=1} \beta_{j+1}(-\omega + |\Delta|)c_{2j-1} \tag{1.46}$$
$$= 0. \tag{1.47}$$

Where we used $\{c_m c_n, c_l\} = 2(\delta_{ln} c_m - \delta_{lm} c_n)$, crossed out terms where $\delta_{l+1,j}$ will always be 0, and redefined $j = j + 1$ in the c_{2j+2} term to make all terms c_{2j}. These terms are general. In the bulk, the three sums can be combined into one as we can ignore the boundary cases $j = 1$ and $j = L$. And at the boundaries one of the three terms will be irrelevant (as we would have a missing site), but the statement $[H, b'] = [H, b''] = 0$ still holds true and gives the boundary conditions. Our problem now reduces to finding β_j that are localized near a boundary and satisfy the above equations. Looking at the bulk we see that we can try solutions on the form $\beta'_j = x^j, \beta''_j = x^{-j}$, and that the equations then will reduce to the same characteristic equation

$$(\omega + |\Delta|)x^{j+1} + \mu x^j + (\omega - |\Delta|)x^{j-1} = 0 \tag{1.48}$$
$$(\omega + |\Delta|)x^2 + \mu x + (\omega - |\Delta|) = 0 \tag{1.49}$$
$$\rightarrow x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4|\Delta|^2}}{2(\omega + |\Delta|)}. \tag{1.50}$$
$$\tag{1.51}$$

From this the general solutions follow on the form

$$\beta'_j = \alpha'_+ x_+^j + \alpha'_- x_-^j, \tag{1.52}$$
$$\beta''_j = \alpha''_+ x_+^{-j} + \alpha''_- x_-^{-j}. \tag{1.53}$$

With $\alpha_\pm$ being constants related to $x_\pm$. Therefore we can write an ansatz for the zero modes on either side of the chain

$$b' = \sum_j \left(\alpha'_+ x_+^j + \alpha'_- x_-^j\right) c_{2j-1} \tag{1.54}$$

$$b'' = \sum_j \left(\alpha''_+ x_+^{-j} + \alpha''_- x_-^{-j}\right) c_{2j} \tag{1.55}$$

We have previously confirmed that the topological phase boundary lies at $2|\omega| = |\mu|, \Delta \neq 0$, with expected majoranas at (a) $2|\omega| > |\mu|, \Delta \neq 0$, and no expected majoranas at (b) $2|\omega| < |\mu|, \Delta \neq 0$. We therefore want to check the boundary modes of these two phases to ensure that this is indeed correct, and we remain in the expected topological phase within these domains.

Before doing so we must clarify our expectations. If the zero mode b' denotes a zero mode on the left side of the chain, then b'' will denote a zero mode on the right side (or vice versa). This is due to the exponential form x^j: If $|x_\pm| < 1$, then $x_\pm^j \to 0$ as j increases, causing the zero mode b' to be localized at the first site, $j = 1$. Whereas $x_\pm^{-j} \to \infty$ as j increases, which after normalization causes b'' to be localized at the end of the chain (again vice versa for $|x_\pm| > 1$). As such, we expect that when both $|x_\pm| < 1$ or $|x_\pm| > 1$ we have localized zero modes, with the second case causing them to have opposite orientations.

If however one solution is smaller than 1, and the other is larger, $|x_\pm| < 1, |x_\mp| > 1$, we see that the zero modes b', b'' become a sum of one element approaching 0, and another approaching ∞. In other words, one component is 'localized' at an end, while the other is growing into the bulk. This is not a localized zero mode. If this were to produce a localized zero mode at one end of the chain, we'd need one of the $\alpha_\pm = 0$. This however breaks with boundary conditions. If we check the simplest missing state at the end of a chain $j = 0$ for b' we see that $\beta'_0 = \alpha'_+ + \alpha'_- = 0$, which is impossible to satisfy if one $\alpha_\pm = 0$, and the other $\alpha_\mp \neq 0$. Similar arguments can be made for both ends of the chain for both b', b''. As such, we expect the case $|x_\pm| < 1, |x_\mp| > 1$ to produce no localized zero modes.

Figuring out when $|x_\pm|$ is larger than or smaller than 1 is not directly obvious, but it can done with a mathematical trick. If we cast $|x_\pm|$ into the complex plane and setting $|x_\pm| = 1 = e^{i\theta}$ to be the unit sphere, we can figure out the conditions on the parameters when $|x_\pm|$ crosses the unit sphere. Plugging this into our characteristic equation 1.49 we find

$$(\omega + |\Delta|)\, e^{i\theta} + \mu + (\omega - |\Delta|)\, e^{-i\theta} = 0 \tag{1.56}$$

$$2\omega \cos(\theta) + 2i|\Delta| \sin(\theta) + \mu = 0 \tag{1.57}$$

Where the complex and the real terms have to equal 0 separately, meaning we require $\sin(\theta) = 0 \to \theta = 0, \pi$, giving the solution

$$2\omega = \pm\mu \tag{1.58}$$

$$2|\omega| = |\mu|. \tag{1.59}$$

Meaning that the value of $|x_\pm|$ can only cross 1 when this condition is met[4]. As such, we can study the domains above and below this condition to determine which topological phase the domain belongs to, and we will remain in this phase if this boundary is not crossed.

[4]It is perhaps unsurprising and definitely reassuring that this is the same result as earlier

(a) $2|\omega| < |\mu|, \Delta \neq 0$

From our condition above we see that we can safely consider the case $\omega = |\Delta|$, which significantly simplifies the problem

$$x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4\omega^2}}{4\omega} \tag{1.60}$$

$$= \frac{-\mu \pm \mu}{4\omega} \tag{1.61}$$

$$\rightarrow x_- = \frac{-\mu}{2\omega}, \quad x_+ = 0 \tag{1.62}$$

$$\rightarrow |x_-| < 1, \quad |x_+| > 1. \tag{1.63}$$

If we were to use different signs for the variables the inequalities would simple be reversed, but never the same for both $|x_\pm|$. As seen from our discussion above this corresponds to the zero modes not existing in this case.

(b) $2\omega > |\mu|, \Delta \neq 0$

If we again consider $\omega = |\Delta|$ we find

$$x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4\omega^2}}{4\omega} \tag{1.64}$$

$$= \frac{-\mu \pm \mu}{4\omega} \tag{1.65}$$

$$\rightarrow x_- = \frac{-\mu}{2\omega}, \quad x_- = \frac{0}{2\omega} \tag{1.66}$$

$$\rightarrow |x_-| < 1, \quad |x_+| < 1. \tag{1.67}$$

In this case we have Majorana zero modes b' localized near $j = 0$, and b'' near $j = L$. And if we change the sign convenction to $-\omega > |\mu|$, we simply flip the position of b' and b''. This confirms our initial expectation that the zero modes will remain in the system when the parameters are changed, as long as $|\Delta| \neq 0$ and we do not cross the boundary from the domain $2|\omega| > |\mu|$ to $2|\omega| < |\mu|$.

1.2.3 Topological phase diagram

From this we get the plot of topological values in Eivinds(?)

1.2.4 Overlap of the Majorana wave functions

Up until this point, the discussion has really been for either a periodic chain, or a chain with $L \rightarrow \infty$. But in a realistic system, we'd have a finite L.

Previously, when we said the Majorana zero modes were localized at an end of the chain, we said so because the coefficient of the majorana operator in b', b'' we either falling off or increasing exponentially with j. This means that they are not truly localized, but have a tail which falls off exponentially from their respective end of the chain.

1.3 BdG

1.3.1 General BdG form and Nambu basis

"the objective of the transformation is to cast a quadratic Hamiltonian into a diagonal form that resembles the Hamiltonian of a system of non-interacting particles." - [5]

1.3.2 BdG form of the Kitaev chain Hamiltonian

We want to express the Hamiltonian in momentum space as

$$H = \frac{1}{2}\sum_q \Psi_q^\dagger \mathcal{H}_{\mathrm{BdG}} \Psi_q, \tag{1.68}$$

using the Nambu basis (spinors)

$$\Psi_q^\dagger = \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix}, \quad \Psi_q = \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}. \tag{1.69}$$

Where the momentum-space fermioninic annihilation and creation operators $a_q, a_q^\dagger$ are defined by the fourier transformation

$$a_j = \frac{1}{\sqrt{N}}\sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}}\sum_q e^{-iqj} a_q^\dagger. \tag{1.70}$$

Where $q = \frac{2\pi n}{N}$ with n such that q runs over the Brillouin zone $[-\pi, \pi]$. We see that $\mathcal{H}_{\mathrm{BdG}}$ has to be a 2×2 matrix. If we look at an arbitrary $\mathcal{H}_{\mathrm{BdG}}$ we can see that after multiplication with the Nambu spinors the Hamiltonian becomes

$$H = \frac{1}{2}\sum_q \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix} \begin{pmatrix} a(q) & b(q) \\ c(q) & d(q) \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix} \tag{1.71}$$

$$= \frac{1}{2}\sum_q \left(a(q) a_q^\dagger a_q + b(q) a_q^\dagger a_{-q}^\dagger + c(q) a_{-q} a_q + d(q) a_{-q} a_{-q}^\dagger \right). \tag{1.72}$$

This should match the momentum-space transformed Hamiltonian as found in A.2. We therefore rewrite the Hamiltonian to match the above description. We use the anti-commutation relations of the operators together with $q = -q$ from the periodic boundary conditions, and drop inessential constants

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega\cos(q) a_q^\dagger a_q - i\sin(q)\left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{1.73}$$

$$= \frac{1}{2}\sum_q \left[2\left(-\mu - 2\omega\cos(q) \right) a_q^\dagger a_q - 2i\sin(q)\left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{1.74}$$

$$= \frac{1}{2}\sum_q \left[\left(-\mu - 2\omega\cos(q) \right) a_q^\dagger a_q + \left(-\mu - 2\omega\cos(q) \right) a_q^\dagger a_q - 2i\sin(q)\left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{1.75}$$

$$= \frac{1}{2}\sum_q \left[\left(-\mu - 2\omega\cos(q) \right) a_q^\dagger a_q + \left(\mu + 2\omega\cos(-q) \right) a_{-q} a_{-q}^\dagger - 2i\sin(q)\left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right]. \tag{1.76}$$

We see that we must have

$$a(q) = -\mu - 2\omega\cos(q) \tag{1.77}$$

$$b(q) = -2i\Delta\sin(q) \tag{1.78}$$

$$c(q) = 2i\Delta^*\sin(q) \tag{1.79}$$

$$d(q) = \mu + 2\omega\cos(-q) = \mu + 2\omega\cos(q) \tag{1.80}$$

$$\rightarrow \mathcal{H}_{\mathrm{BdG}} = \begin{pmatrix} -\mu - 2\omega\cos(q) & -2i\Delta\sin(q) \\ 2i\Delta^*\sin(q) & \mu + 2\omega\cos(q) \end{pmatrix}. \tag{1.81}$$

Diagonalizing the BdG Hamiltonian

By diagonalizing the BdG Hamiltonian we can find the energy spectrum. We find its eigenvalues by

$$\det\left(\mathcal{H}_{\mathrm{BdG}} - \epsilon I\right) = 0 \tag{1.82}$$
$$\tag{1.83}$$

which by the abc-formula gives the energy spectrum

$$\epsilon = \pm\sqrt{\left(\mu + 2\omega\cos(q)\right)^2 + 4|\Delta|^2\sin^2(q)} \tag{1.84}$$

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Appendix A

Calculations

A.1 Transforming the Kitaev chain Hamiltonian to Majorana operators

This appendix shows how transform the original Kitaev chain Hamiltonian expressed in fermionic operators a_j and $a_j^\dagger$ to being expressed in terms of the majorana operators c_{2j-1} and $c_2 j$.

Starting with the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{A.1}$$
$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{A.2}$$

We can invert them, leading to four expressions

$$a_j = e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger \tag{A.3}$$
$$= ie^{-i\frac{\theta}{2}} c_{2j} + e^{-i\theta} a_j^\dagger \tag{A.4}$$
$$a_j^\dagger = e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j \tag{A.5}$$
$$= -ie^{i\frac{\theta}{2}} c_{2j} + e^{i\theta} a_j. \tag{A.6}$$

Combining the expressions for a_j, and for $a_j^\dagger$ we find

$$a_j = \frac{1}{2}\left(e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger + e^{-i\theta} a_j^\dagger + ie^{-i\frac{\theta}{2}} c_2 j\right) \tag{A.7}$$
$$= \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + ic_{2j}\right) \tag{A.8}$$
$$a_j^\dagger = \frac{1}{2}\left(e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j + e^{i\theta} a_j - ie^{i\frac{\theta}{2}} c_{2j}\right) \tag{A.9}$$
$$= \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - ic_{2j}\right). \tag{A.10}$$

We now substitute these expressions into the original Hamiltonian

$$
\begin{aligned}
H =& \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] && \text{(A.11)} \\
=& -\omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) && \text{(A.12)} \\
& - \omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) && \text{(A.13)} \\
& - \mu \sum_j \left(\frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) - \frac{1}{2} \right) && \text{(A.14)} \\
& + \Delta \sum_j \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) && \text{(A.15)} \\
& + \Delta^* \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) && \text{(A.16)} \\
=& -\frac{\omega}{4} \sum_j \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} - i c_{2j} c_{2j+1} + c_{2j} c_{2j+2} \right) && \text{(A.17)} \\
& - \frac{\omega}{4} \sum_j \left(c_{2j+1} c_{2j-1} + i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} + c_{2j+2} c_{2j} \right) && \text{(A.18)} \\
& - \frac{\mu}{4} \sum_j \left(c_{2j-1}^2 + i c_{2j-1} c_{2j} - i c_{2j} c_{2j-1} + c_{2j}^2 - 2 \right) && \text{(A.19)} \\
& + \frac{|\Delta|}{4} e^{i\theta} \sum_j e^{-i\theta} \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} + i c_{2j} c_{2j+1} - c_{2j} c_{2j+2} \right) && \text{(A.20)} \\
& + \frac{|\Delta|}{4} e^{-i\theta} \sum_j e^{i\theta} \left(c_{2j+1} c_{2j-1} - i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} - c_{2j+2} c_{2j} \right) && \text{(A.21)} \\
=& -\frac{\omega}{4} \sum_j \left(0 + 2i c_{2j-1} c_{2j+2} - 2i c_{2j} c_{2j+1} + 0 \right) && \text{(A.22)} \\
& - \frac{\mu}{4} \sum_j \left(2i c_{2j-1} c_{2j} + 2 - 2 \right) && \text{(A.23)} \\
& + \frac{|\Delta|}{4} \sum_j \left(0 + 2i c_{2j-1} c_{2j+2} + 2i c_{2j} c_{2j+1} + 0 \right) && \text{(A.24)} \\
=& \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + \left(\omega + |\Delta| \right) c_{2j} c_{2j+1} + \left(-\omega + |\Delta| \right) c_{2j-1} c_{2j+2} \right] && \text{(A.25)}
\end{aligned}
$$

Where we used $\Delta = |\Delta| e^{i\theta}$ and the anticommutation relations

$$
\begin{aligned}
\{ c_l, c_m \} = c_l c_m + c_m c_l &= 2\delta_{lm} && \text{(A.26)} \\
c_l c_m &= -c_m c_l (m \neq l) && \text{(A.27)} \\
c_l^2 &= 1. && \text{(A.28)}
\end{aligned}
$$

A.2 Transforming the Kitaev chain Hamiltonian to momentum space

Taking inspiration from [4] we start from the Kitaev chain Hamiltonian

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{A.29}$$

Imposing periodic boundary conditions we can Fourier transform the Hamiltonian to momentum space q. We recall the Fourier transform

$$a_j = \frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \tag{A.30}$$

where $q = \frac{2\pi n}{N}$, with n such that $q$runs over the first Brillouin zone $[-\pi, \pi]$. These operators keep the anticommutation relations of the real-space operators

$$\{a_q, a_{q'}\} = \frac{1}{N} \sum_{jj'} e^{-iqj} e^{-iq'j'} \{a_j, a_{j'}\} = 0 \tag{A.31}$$

$$\left\{a_q, a_{q'}^\dagger\right\} = \frac{1}{N} \sum_{jj'} e^{-iqj} e^{iq'j'} \left\{a_j, a_{j'}^\dagger\right\} \tag{A.32}$$

$$= \frac{1}{N} \sum_{jj'} e^{-iqj} e^{iq'j'} \delta_{j,j'} \tag{A.33}$$

$$= \frac{1}{N} \sum_{j} e^{i(q'-q)j} \tag{A.34}$$

$$= \delta_{q,q'} \tag{A.35}$$

Using this we can transform each term in the Hamiltonian individually for simplicity. Starting with the on-site term

$$H_\mu = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) \tag{A.36}$$

$$= -\mu \sum_j \left(\left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'j} a_{q'} \right) - \frac{1}{2} \right) \tag{A.37}$$

$$= -\mu \frac{1}{N} \sum_{jqq'} e^{i(q'-q)j} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.38}$$

$$= -\frac{\mu}{N} \sum_{qq'} N \delta_{q,q'} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.39}$$

$$= -\mu \sum_q a_q^\dagger a_q + \frac{\mu N}{2}. \tag{A.40}$$

Where we used a property of the periodic boundary conditions $\sum_j e^{i(q'-q)j} = N\delta_{q,q'}$. Continuing with the hopping term

$$H_\omega = -\omega \sum_j \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) \tag{A.41}$$

where we again split the calculation of the terms; the first term becomes

$$H_\omega^1 = -\omega \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'} \right) \tag{A.42}$$

$$= -\frac{\omega}{N} \sum_{jqq'} e^{i(q'-q)j} e^{iq'} a_q^\dagger a_{q'} \tag{A.43}$$

$$= -\frac{\omega}{N} \sum_{qq'} N \delta_{q,q'} e^{iq'} a_q^\dagger a_{q'} \tag{A.44}$$

$$= -\omega \sum_q e^{iq} a_q^\dagger a_q. \tag{A.45}$$

And the second term similarly becomes

$$H_\omega^2 = -\omega \sum_q e^{-iq} a_q^\dagger a_q. \tag{A.46}$$

Combining the hopping terms we get

$$H_\omega = H_\omega^1 + H_\omega^2 \tag{A.47}$$

$$= -\omega \sum_q \left(e^{iq} + e^{-iq} \right) a_q^\dagger a_q \tag{A.48}$$

$$= -2\omega \sum_q \cos(q) a_q^\dagger a_q. \tag{A.49}$$

We now consider the pairing term

$$H_\Delta = \sum_j \left(\Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right) \tag{A.50}$$

$$\tag{A.51}$$

where we again split the calculation for simplicity

$$H_\Delta^1 = \Delta \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'} \right) \tag{A.52}$$

$$= \frac{\Delta}{N} \sum_{jqq'} e^{i(q+q')j} e^{iq'} a_q a_{q'} \tag{A.53}$$

$$= \frac{\Delta}{N} \sum_{qq'} N \delta_{q,-q'} e^{iq'} a_q a_{q'} \tag{A.54}$$

$$= \Delta \sum_q e^{-iq} a_q a_{-q}. \tag{A.55}$$

And the second term similarly becomes

$$H_\Delta^2 = \Delta^* \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{-iq(j+1)} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{-iq'j} a_{q'}^\dagger \right) \tag{A.56}$$

$$= \frac{\Delta^*}{N} \sum_{jqq'} e^{-i(q+q')j} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.57}$$

$$= \frac{\Delta^*}{N} \sum_{qq'} N \delta_{q,-q'} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.58}$$

$$= \Delta^* \sum_q e^{-iq} a_q^\dagger a_{-q}^\dagger. \tag{A.59}$$

Combining all terms we get the momentum-space Hamiltonian

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q + \Delta e^{-iq} a_q a_{-q} + \Delta^* e^{-iq} a_q^\dagger a_{-q}^\dagger \right] + \frac{\mu N}{2} \tag{A.60}$$

Where the last term $+\frac{\mu N}{2}$ can be ignored as it does not affect the spectrum, eigenstates, or any observable that depends on the energy differences. As such it will be disregarded for the rest of the thesis. Later we will use this expression to construct the BdG Hamiltonian, which requires the pairing terms to be in an antisymmetric form, which they are not currently in. We can create the antisymmetric form by expanding the operators to be a combination of both orderings $a_q a_{-q}, a_{-q} a_q$ (and similar for the † terms) by the anticommutation relations. We start by considering the annihilation term

$$\sum_q e^{-iq} a_q a_{-q} = \frac{1}{2} \sum_q \left[e^{-iq} a_q a_{-q} + e^{-iq} a_q a_{-q} \right] \tag{A.61}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_q e^{-iq} a_{-q} a_q \right] \tag{A.62}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_{-q} e^{iq} a_q a_{-q} \right] \tag{A.63}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_q e^{iq} a_q a_{-q} \right] \tag{A.64}$$

$$= \frac{1}{2} \sum_q \left[e^{-iq} - e^{iq} \right] a_q a_{-q} \tag{A.65}$$

$$= \frac{1}{2} \sum_q \left[-2i \sin(q) \right] a_q a_{-q} \tag{A.66}$$

$$= -i \sum_q \sin(q) a_q a_{-q} \tag{A.67}$$

$$\tag{A.68}$$

Where we also renambed the dummy variable $q \to -q$ for the second term. We are allowed to do this as a sum over q is equivalent to a sum over $-q$ by the periodic boundary conditions. The corresponding calculation for the creation term similarly gives

$$\sum_q e^{-iq} a_q^\dagger a_{-q}^\dagger = -i \sum_q \sin(q) a_q^\dagger a_{-q}^\dagger. \tag{A.69}$$

As such the final momentum-space Hamiltonian with antisymmetric pairing terms is

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q - i\Delta \sin(q) a_q a_{-q} - i\Delta^* \sin(q) a_q^\dagger a_{-q}^\dagger \right] \tag{A.70}$$

$$= \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q - i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{A.71}$$

A.3 Finding the evolution of the system under the adiabatic assumption

Move huge ugly calculation from methods to here(?)